Anatomy

Q: What is the order of meningeal layers, from outer to inner?
Answer: Epidural space, Dura mater, Subdural space, Arachnoid mater, Subarachnoid space, Pia mater.

Q: What is the difference between Cona Medularis injury and Cauda Equina injury?
Answer: Cauda Equina has + babinski sign

Q: In one of the cases, the patient had flattened forehead furrow, because of?
Answer: Facial nerve lesion

Q: A case of patient lost sensation on ipsilateral side of the face and contralateral side of the whole body.
Answer: Lateral medullary syndrome

Q: Which branches of spinal nerves make plexuses?
Answer: Anterior rami

Case:
Patient has right loss of sensory and motor in face. Left deafness and vertigo. Right internal stubismus
Q: Where is the lesion? **A:** Pons
Q: Why there is deafness? **A:** Involvement of left lateral laminiscus
Q: Why there is stabismus? **A:** Due to involvement of abducent nerve

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Biochemistry

Q: How is glutamate inactivated as a neurotransmitter?
Answer: By uptake into astrocytes and conversion to glutamine

Q: Glutamate can be formed from which of the krebs cycle intermediates?
Answer: Alpha-ketoglutarate

Q: Because of thrombosis, penumbra will die and core of the infarct increases because...?
Answer: Increased Utilization of ATP

Q: Mechanism by which when membrane depolarized to the highest extent and glutamate is effluxed...- NT uptake is...?
Answer: Na dependent mechanism

Q: Closure of K channel in nerve fibers lead to?
Answer: Metabotropic ESPS.

Clinical

Which of the following kinds is not generalized epilepsy?
a. Tonic-clonic
b. Atonic
c. Myoclonic
d. Jacksonian
e. Clonic

What is the highest level of consciousness eye grading according to the Glasgow Coma scale?
a. 1
b. 2
c. 3
d. 4
e. 5

What is the position of a decerebrate posture?
a. Adducted arm, flexed elbow, flexed wrist, extended knee, and plantar flexion
b. Adducted arm, extended elbow, pronated lower arm, flexed elbow, extended knee, and plantar flexion
c. Abducted arm, extended elbow, flexed knee
d. Flexed knee, plantar flexion, flexed elbow, pronated lower arm, extended elbow, adducted arm
e. Adducted arm, flexed elbow, flexed, wrist, extended knee, and dorsiflexion

What is the pattern Chynes-Stokes ventilation?
Answer: Hyperventilation, and apnea

Seizure occurs due to the increase activity of ……? > gray mater

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Microbiology

The route of transmission of polio virus is?
a. Droplet transmission
b. Animal bite
c. Fecal-oral
d. Sheep
e. Birds' dropping

The reactivation of varicella-zoster virus causes?
Answer: Shingles

Picture with strep agalactia...
Picture with nissera asking what u see in CSF
girl from India came with low grade fever 3 weeks ago, csf: low glucose, high protein, likely finding on meninges?
- Streptococcus pneumoniae virulence factor? capsule, Ig protease
- a case with symptoms of GBS and gastroenteritis, what is the causative agent? Campylobacter jejuni
Encephalitis transmitted by feco-oral rote > Polio virus

Pharmacology

Which of the following drugs is contraindicated in pregnancy?
a. Beta-lactams
b. Penicillin
c. Carbapenems
d. Fluoroquinolones
e. Cephalosporins

Which of the pain grading scales is used for chronic pain?
a. Multi-dimensional
b. Numeric
c. Visual analog
d. Visual
e. Single dimensional

Which of the following opioids has the highest affinity to μ (mu) receptors?
a. Endorphins
b. Met-Enkaphalins
c. Dynorphins
d. B-dynorphin
e. Ephedrine

Which beta-blocker has established efficacy in the prevention of migraines?
a. Atenolol
b. Propranolol
c. Esmolol
d. Labetalol
e. Carvedilol

What is the first line treatment for streptococcus pneumonia?
Answer: Vancomycin and cefotaxime
Thiopental has a short half-life due to?
Answer: It rapidly redistributes out of the brain.
Esters have shorter half-life compared to amides due to the action of?
Answer: Plasma enzymes
Which of the following is a contraindication for abortive therapy in migraines? - **angina**
which opioid do you give man with chronic severe back pain with SSRI? **Morphine**
Suppression of pain - periaquductal grey
How do opioids act presynaptically
What do we use for chronic pain assessment? **McGrill scale**

Pathology

A 35-year old woman has noticed marked pain in the right foot for the past 2 months. The pain makes it difficult for her to wear high-heeled shoes and seems to be worse at the end of the day. On physical examination, she has severe pain on palpation of the inter-digital space between the second and third toes. There is no swelling or erythema of the foot. Motor strength in the lower extremities seems to be normal. What has most likely produced these findings?

a. Diabetes mellitus types 2

b. Entrapment neuropathy

c. Lead poisoning

d. Thiamine deficiency

e. Neoplastic proliferation of Schwann cells

A 58-year old man has an episode of hemoptysis. Over the next week, he becomes progressively obtunded. He has a blood pressure of 110/170 mmHg, pulse 90/minute, respirations 25/minute, and temperature 36.8 C. A chest x-ray shows a right hilar lung mass. Serum chestries reveal a sodium of 108mmol/L, potassium 3.9mmol/L, and chloride 85 mmol/L. He is given intravenous saline, and the serum sodium is 135mmol/L the next day. He then becomes comatose and dies. On autopsy, which of the following lesions is most likely to be found in the brain?

a. Metastases to the cerebral hemispheres

b. Central pontine myelinolysis

c. Acute hisseminated encephalomyelitis

d. HIV leukoencephalopathy

e. Multiple sclerosis

- enlarged oligodendrocytes contain huge eosinophilic intranuclear inclusions in? **PML**
- cowdry bodies >>> ? **herpes virus**
- cytotoxic increase in CSF indicates > **infarction**
- Patient had car accident and died after 3 months. Microscopic examination of the brain will reveal which type of glial cells > **astrocytes**
- Patient did lumbar puncture. It showed 300 lymphocytes, glucose 2/3 of plasma glucose and increase protien. What is the next step? **Enzyme assay**.

Physiology

What does the muscle spindle tell the CNS about the muscle

You and your friend are recording the velocity of fibers, the nerve conduction velocity shows 100m/s, what type of fiber is this?

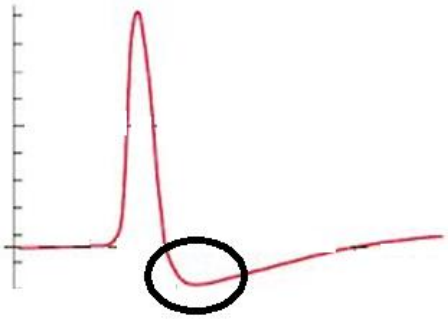
a. A alpha

b. A beta

c. wC

d. A gamma

Which ion is responsible for the marked area in the following figure?



a. Na

b. K

c. Cl

d. Mg

e. NaCl

A person is awake and relaxed, which type of wave would an EEG record?

a. Delta

b. Theta

c. Beta

d. Alpha

e. Gamma

Which of the following sites has no blood brain barrier?

Neurohypophysis

There resting membrane potential of a cell is caused by?

a. Different membrane permeability of ions

b. Magnesium

c. The presence of non-diffusable negative molecules in the cell

d. The presence of non-diffusable negative molecules outside the cell

e. The sodium potassium ATPase

- type of NT that RAS has > **serotonin**
- CGRP function > **vasodilation**
- if you cut afferent from muscle spindle what happens - **hypotonia**
- Which brainwave do we see when an adult is awake but relaxed? **Alpha waves**
- What causes this on the graph (graph showed relative refractory period) > **open K channels**
- the nerst equation
- What type of neuronal circuit is used for rhythmic activities? **Central pattern generators**

- in hydrocephalus case, the lateral ventricle was found to be enlarged, where is the blockage? **foramen of monro**
- axon diameter increases its velocity
- closing K+ channels? **Slow EPSP**
- muscle spindle detects? **muscle length**
- contraction of the flexors and inhibition of extensors with extension of the opposite limb is called? **withdrawal reflex**
- receptor is excited most effectively by only one stimulus known as? **adequate stimulus**
- Which part of brain inhibits pain? **Periqueductal gray mater**.
- Afferent sensory fibers is transmitted by? **Golgi**
- Dreaming occurs during ……. > **REM sleep**

Best Of Luck !!